Capstone R Work

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This code retrieves data from both the World Bank and UNDB databases via API calls. The APi calls are in python and therefore are run using the reticulate package which allows seamless exchange between R and Python. Once the data is cleaned and joined into a single dataframe, that dataframe is written to a .rds file for use in the EDA script

```r
library(reticulate)
library(tidyverse)
```

```
## -- Attaching core tidyverse packages ------------------------ tidyverse 2.0.0 --
## v dplyr     1.1.4     v readr     2.1.5
## v forcats   1.0.0     v stringr   1.5.1
## v ggplot2   3.5.2     v tibble    3.3.0
## v lubridate 1.9.4     v tidyr     1.3.1
## v purrr     1.1.0
## -- Conflicts ------------------------------------------ tidyverse_conflicts() --
## x dplyr::filter() masks stats::filter()
## x dplyr::lag()    masks stats::lag()
## i Use the conflicted package (<http://conflicted.r-lib.org/>) to force all conflicts to become errors
```

```r
library(ggplot2)
library(dplyr)
library(here)
```

```
## here() starts at /Users/raychandonnet/Chandonnet Family Dropbox/Chandonnet Family Folder/Ray Personal
```

```r
# Portable setup for Python environment via reticulate
if (!requireNamespace("reticulate", quietly = TRUE)) {
  install.packages("reticulate")
}
library(reticulate)

env_name <- "retic_env2"

# Only create the environment if it doesn't exist
if (!env_name %in% virtualenv_list()) {
  message("Creating virtual environment '", env_name, "' and installing packages...")
  virtualenv_create(
    envname = env_name
  )

  # Install required packages after creation
  py_install(
```

```
    packages = c("pandas", "requests", "openpyxl", "pycountry", "wbgapi"),
    envname = env_name,
    pip = TRUE
  )
} else {
  message("Virtual environment '", env_name, "' already exists.")
}
```

```
## Virtual environment 'retic_env2' already exists.
```

```
# Activate the environment
use_virtualenv(env_name, required = TRUE)

# Confirm setup
py_config()
```

```
## python:         /Users/raychandonnet/.virtualenvs/retic_env2/bin/python
## libpython:      /opt/homebrew/opt/python@3.13/Frameworks/Python.framework/Versions/3.13/lib/python3.
## pythonhome:     /Users/raychandonnet/.virtualenvs/retic_env2:/Users/raychandonnet/.virtualenvs/retic_
## version:        3.13.5 (main, Jun 11 2025, 15:36:57) [Clang 17.0.0 (clang-1700.0.13.3)]
## numpy:          /Users/raychandonnet/.virtualenvs/retic_env2/lib/python3.13/site-packages/numpy
## numpy_version:  2.3.2
##
## NOTE: Python version was forced by use_python() function
```

```
# Turn the python code into a string for Reticulate and import the python libraries needed - BJV
wb_undp_api_python_code <- "
import pandas as pd
import wbgapi as wb
import requests
import openpyxl
import pycountry

wb_series_codes = [
    'NY.GDP.PCAP.KD.ZG',
    'NY.GNP.PCAP.KD.ZG',
    'SE.XPD.TOTL.GD.ZS',
    'SE.ADT.LITR.ZS',
    'SE.COM.DURS',
    'SE.LPV.PRIM',
    'SE.LPV.PRIM.SD',
    'SE.SEC.CUAT.LO.ZS',
    'SE.SEC.CUAT.PO.ZS',
    'SE.SEC.CUAT.UP.ZS',
    'SE.TER.CUAT.BA.ZS',
    'SE.TER.CUAT.DO.ZS',
    'SE.TER.CUAT.MS.ZS',
    'SE.TER.CUAT.ST.ZS',
    'AG.PRD.FOOD.XD',
    'EN.POP.DNST',
    'EN.POP.SLUM.UR.ZS',
    'SP.RUR.TOTL.ZG',
    'EG.ELC.ACCS.ZS',
    'ER.H2O.FWST.ZS',
    'FX.OWN.TOTL.ZS',
```

```
    'SN.ITK.MSFI.ZS',
    'SH.XPD.CHEX.PC.CD',
    'SH.XPD.GHED.PC.CD',
    'SH.STA.WASH.P5',
    'SP.DYN.LE00.IN',
    'SH.UHC.SRVS.CV.XD',
    'IT.NET.BBND.P2',
    'IT.NET.USER.ZS',
    'GB.XPD.RSDV.GD.ZS',
    'IS.SHP.GCNW.XQ',
    'SI.POV.GINI',
    'VC.IHR.PSRC.P5',
    'CC.EST',
    'GE.EST',
    'PV.EST',
    'RL.EST',
    'VA.EST',
    'SM.POP.TOTL.ZS',
    'SI.POV.GINI',
    'NY.GDP.MKTP.KD.ZG',
    'NY.GDP.MKTP.KD'
]

# Create a data frame from the World Bank API - BJV

wb_df = wb.data.DataFrame(
    wb_series_codes,
    economy='all',
    time=range(1995, 2024),
    columns='series'
)

# Index needs to be reset in ordert for 'economy' and 'time' to be accessed and renamed - BJV
wb_df = wb_df.reset_index()

wb_df = wb_df.rename(columns={'economy': 'ISO3', 'time': 'year'})

# Remove YR from the 'year' column to leave just four digit years - BJV

wb_df['year'] = wb_df['year'].astype(str).str.replace('YR', '', regex=False)

# convert the year to a numeric from string for errors arising later - BJV
wb_df['year'] = pd.to_numeric(wb_df['year'], errors='coerce').astype('Int64')


# UNDP API Key and Request - BJV

undp_url = 'https://hdrdata.org/api/CompositeIndices/query?apikey=HDR-KOgLmhxaLjacLvGRi4oTPARBLx54ubNe&

response = requests.get(undp_url)

if response.status_code == 200:
    data = response.json()
```

```
df1 = pd.DataFrame(data)

# UNDP API Key and Request - BJV

undp_url2 = 'https://hdrdata.org/api/CompositeIndices/query?apikey=HDR-KOgLmhxaLjacLvGRi4oTPARBLx54ubNe

response2 = requests.get(undp_url2)

if response2.status_code == 200:
    data2 = response2.json()

df2 = pd.DataFrame(data2)

# UNDP API Key and Request - BJV

undp_url3 = 'https://hdrdata.org/api/CompositeIndices/query?apikey=HDR-KOgLmhxaLjacLvGRi4oTPARBLx54ubNe

response3 = requests.get(undp_url3)

if response3.status_code == 200:
    data3 = response3.json()

df3 = pd.DataFrame(data3)

# UNDP API Key and Request - BJV

undp_url4 = 'https://hdrdata.org/api/CompositeIndices/query?apikey=HDR-KOgLmhxaLjacLvGRi4oTPARBLx54ubNe

response4 = requests.get(undp_url4)

if response4.status_code == 200:
    data4 = response4.json()

df4 = pd.DataFrame(data4)

# UNDP API Key and Request - BJV

undp_url5 = 'https://hdrdata.org/api/CompositeIndices/query?apikey=HDR-KOgLmhxaLjacLvGRi4oTPARBLx54ubNe

response5 = requests.get(undp_url5)

if response5.status_code == 200:
    data5 = response5.json()

df5 = pd.DataFrame(data5)

# UNDP API Key and Request - BJV

undp_url6 = 'https://hdrdata.org/api/CompositeIndices/query?apikey=HDR-KOgLmhxaLjacLvGRi4oTPARBLx54ubNe

response6 = requests.get(undp_url6)

if response6.status_code == 200:
```

```
    data6 = response6.json()

df6 = pd.DataFrame(data6)

# UNDP API Key and Request - BJV

undp_url7 = 'https://hdrdata.org/api/CompositeIndices/query?apikey=HDR-KOgLmhxaLjacLvGRi4oTPARBLx54ubNe&

response7 = requests.get(undp_url7)

if response7.status_code == 200:
    data7 = response7.json()

df7 = pd.DataFrame(data7)

# UNDP API Key and Request - BJV

undp_url8 = 'https://hdrdata.org/api/CompositeIndices/query?apikey=HDR-KOgLmhxaLjacLvGRi4oTPARBLx54ubNe&

response8 = requests.get(undp_url8)

if response8.status_code == 200:
    data8 = response8.json()

df8 = pd.DataFrame(data8)

# UNDP API Key and Request - BJV

undp_url9 = 'https://hdrdata.org/api/CompositeIndices/query?apikey=HDR-KOgLmhxaLjacLvGRi4oTPARBLx54ubNe&

response9 = requests.get(undp_url9)

if response9.status_code == 200:
    data9 = response9.json()

df9 = pd.DataFrame(data9)

# UNDP API Key and Request - BJV

undp_url10 = 'https://hdrdata.org/api/CompositeIndices/query?apikey=HDR-KOgLmhxaLjacLvGRi4oTPARBLx54ubNe

response10 = requests.get(undp_url10)

if response10.status_code == 200:
    data10 = response10.json()

df10 = pd.DataFrame(data10)

# UNDP API Key and Request - BJV

undp_url11 = 'https://hdrdata.org/api/CompositeIndices/query?apikey=HDR-KOgLmhxaLjacLvGRi4oTPARBLx54ubNe

response11 = requests.get(undp_url11)
```

```
if response11.status_code == 200:
    data11 = response11.json()

df11 = pd.DataFrame(data11)

# UNDP API Key and Request - BJV

undp_url12 = 'https://hdrdata.org/api/CompositeIndices/query?apikey=HDR-KOgLmhxaLjacLvGRi4oTPARBLx54ubN

response12 = requests.get(undp_url12)

if response12.status_code == 200:
    data12 = response12.json()

df12 = pd.DataFrame(data12)

# UNDP API Key and Request - BJV

undp_url13 = 'https://hdrdata.org/api/CompositeIndices/query?apikey=HDR-KOgLmhxaLjacLvGRi4oTPARBLx54ubN

response13 = requests.get(undp_url13)

if response13.status_code == 200:
    data13 = response13.json()

df13 = pd.DataFrame(data13)

# UNDP API Key and Request - BJV

undp_url14 = 'https://hdrdata.org/api/CompositeIndices/query?apikey=HDR-KOgLmhxaLjacLvGRi4oTPARBLx54ubN

response14 = requests.get(undp_url14)

if response14.status_code == 200:
    data14 = response14.json()

df14 = pd.DataFrame(data14)

# UNDP API Key and Request - BJV

undp_url15 = 'https://hdrdata.org/api/CompositeIndices/query?apikey=HDR-KOgLmhxaLjacLvGRi4oTPARBLx54ubN

response15 = requests.get(undp_url15)

if response15.status_code == 200:
    data15 = response15.json()

df15 = pd.DataFrame(data15)

all_dataframes = [df1, df2, df3, df4, df5, df6, df7, df8,
                  df9, df10, df11, df12, df13, df14, df15]

# Create a list of all data frames to be concatenated. -BJV
```

```
list_of_dfs = [df1, df2, df3, df4, df5, df6, df7, df8,
               df9, df10, df11, df12, df13, df14, df15]

# Concatenate all the data frames in the list. - BJV

combined_df = pd.concat(list_of_dfs, ignore_index=True)


# Create a df from a dictionary for saving file - BJV
undp_full_df = pd.DataFrame(combined_df)

# Country needed to split the ISO3 and the Country names - BJV
split_data = undp_full_df['country'].str.split(' - ', n=1, expand=True)

undp_full_df['ISO3 Code'] = split_data[0]
undp_full_df['Country Name'] = split_data[1]

# Drop the original 'Country' column - BJV
undp_full_df_split = undp_full_df.drop(columns=['country'])

# 'Value' was causing and error and needed to be converted to numeric - BJV
undp_full_df_split['value'] = pd.to_numeric(undp_full_df_split['value'], errors='coerce')

# Data needed to be pivoted for clarity when doing EDA - BJV

undp_full_df_pivoted = undp_full_df_split.pivot_table(index=['ISO3 Code', 'year'],
                            columns='indicator',
                            values='value')

undp_full_df_pivoted = undp_full_df_pivoted.rename(columns={'ISO3 Code': 'ISO3'})


# These are the columns that are relevant to our study from the total column options - BJV

columns_to_keep = [
    'hdi - Human Development Index (value)',
    'gii_rank - GII Rank',
    'rankdiff_hdi_phdi - Difference from HDI rank',
    'ihdi - Inequality-adjusted Human Development Index (value)',
    'coef_ineq - Coefficient of human inequality',
    'ineq_le - Inequality in life expectancy',
    'le - Life Expectancy at Birth (years)',
    'ineq_edu - Inequality in eduation',
    'ineq_inc - Inequality in income',
    'coef_ineq - Coefficient of human inequality'
]

undp_full_df_pivoted.index.rename(
    ['ISO3', 'year'],
    inplace=True
)

undp_df_filtered = undp_full_df_pivoted[columns_to_keep]
```

```
undp_df_filtered = undp_df_filtered.reset_index()

undp_df_filtered['year'] = pd.to_numeric(undp_df_filtered['year'], errors='coerce').astype('Int64')

# Joined Work Bank and UNDP Data Frames - BJV

undp_wb_filtered_full = pd.merge(undp_df_filtered, wb_df, on=['ISO3', 'year'], how='left')

# Use pycountry to map country names to ISO3 codes -BJV
iso3_to_name_map = {}
for country in pycountry.countries:
    iso3_to_name_map[country.alpha_3] = country.name

mapped_full_df = pd.DataFrame(undp_wb_filtered_full)

# Add the 'Country Name' column - BJV
mapped_full_df['Country Name'] = mapped_full_df['ISO3'].map(iso3_to_name_map)

cols = mapped_full_df.columns.tolist()

current_index_of_country_name = cols.index('Country Name')

# Pop country name - BJV

column_to_move = cols.pop(current_index_of_country_name)

# Move 'country name' to column 2
cols.insert(1, column_to_move)

mapped_full_df = mapped_full_df[cols]"


reticulate::py_run_string(wb_undp_api_python_code)

wb_undp_df_r <- py$mapped_full_df
```

```
# Example of the python code turning into a dataframe that works with tidyverse, etc. - BJV
# Also - save the dataframe into an .rds file to be imported into our EDA and then feature engineering
glimpse(wb_undp_df_r)
```

```
## Rows: 5,974
## Columns: 56
## $ ISO3                                                    <chr> "AFG", "A~
## $ `Country Name`                                          <chr> "Afghanis~
## $ year                                                    <int> 1995, 199~
## $ `hdi - Human Development Index (value)`                 <dbl> 0.329, 0.~
## $ `gii_rank - GII Rank`                                   <dbl> NaN, NaN,~
## $ `rankdiff_hdi_phdi - Difference from HDI rank`          <dbl> NaN, NaN,~
## $ `ihdi - Inequality-adjusted Human Development Index (value)` <dbl> NaN, NaN,~
## $ `coef_ineq - Coefficient of human inequality`           <dbl> NaN, NaN,~
## $ `coef_ineq - Coefficient of human inequality`           <dbl> NaN, NaN,~
## $ `ineq_le - Inequality in life expectancy`               <dbl> NaN, NaN,~
## $ `le - Life Expectancy at Birth (years)`                 <dbl> 52.103, 5~
```

```
## $ `ineq_edu - Inequality in eduation`                    <dbl> NaN, NaN,~
## $ `ineq_inc - Inequality in income`                      <dbl> NaN, NaN,~
## $ `coef_ineq - Coefficient of human inequality`          <dbl> NaN, NaN,~
## $ `coef_ineq - Coefficient of human inequality`          <dbl> NaN, NaN,~
## $ AG.PRD.FOOD.XD                                         <dbl> 67.87, 71~
## $ CC.EST                                                 <dbl> NaN, -1.2~
## $ EG.ELC.ACCS.ZS                                         <dbl> NaN, NaN,~
## $ EN.POP.DNST                                            <dbl> 26.16536,~
## $ EN.POP.SLUM.UR.ZS                                      <dbl> NaN, NaN,~
## $ ER.H2O.FWST.ZS                                         <dbl> 59.04386,~
## $ FX.OWN.TOTL.ZS                                         <dbl> NaN, NaN,~
## $ GB.XPD.RSDV.GD.ZS                                      <dbl> NaN, NaN,~
## $ GE.EST                                                 <dbl> NaN, -2.1~
## $ IS.SHP.GCNW.XQ                                         <dbl> NaN, NaN,~
## $ IT.NET.BBND.P2                                         <dbl> NaN, NaN,~
## $ IT.NET.USER.ZS                                         <dbl> NaN, NaN,~
## $ NY.GDP.MKTP.KD                                         <dbl> NaN, NaN,~
## $ NY.GDP.MKTP.KD.ZG                                      <dbl> NaN, NaN,~
## $ NY.GDP.PCAP.KD.ZG                                      <dbl> NaN, NaN,~
## $ NY.GNP.PCAP.KD.ZG                                      <dbl> NaN, NaN,~
## $ PV.EST                                                 <dbl> NaN, -2.4~
## $ RL.EST                                                 <dbl> NaN, -1.7~
## $ SE.ADT.LITR.ZS                                         <dbl> NaN, NaN,~
## $ SE.COM.DURS                                            <dbl> NaN, NaN,~
## $ SE.LPV.PRIM                                            <dbl> NaN, NaN,~
## $ SE.LPV.PRIM.SD                                         <dbl> NaN, NaN,~
## $ SE.SEC.CUAT.LO.ZS                                      <dbl> NaN, NaN,~
## $ SE.SEC.CUAT.PO.ZS                                      <dbl> NaN, NaN,~
## $ SE.SEC.CUAT.UP.ZS                                      <dbl> NaN, NaN,~
## $ SE.TER.CUAT.BA.ZS                                      <dbl> NaN, NaN,~
## $ SE.TER.CUAT.DO.ZS                                      <dbl> NaN, NaN,~
## $ SE.TER.CUAT.MS.ZS                                      <dbl> NaN, NaN,~
## $ SE.TER.CUAT.ST.ZS                                      <dbl> NaN, NaN,~
## $ SE.XPD.TOTL.GD.ZS                                      <dbl> NaN, NaN,~
## $ SH.STA.WASH.P5                                         <dbl> NaN, NaN,~
## $ SH.UHC.SRVS.CV.XD                                      <dbl> NaN, NaN,~
## $ SH.XPD.CHEX.PC.CD                                      <dbl> NaN, NaN,~
## $ SH.XPD.GHED.PC.CD                                      <dbl> NaN, NaN,~
## $ SI.POV.GINI                                            <dbl> NaN, NaN,~
## $ SM.POP.TOTL.ZS                                         <dbl> 0.4264236~
## $ SN.ITK.MSFI.ZS                                         <dbl> NaN, NaN,~
## $ SP.DYN.LE00.IN                                         <dbl> 52.103, 5~
## $ SP.RUR.TOTL.ZG                                         <dbl> 4.7789179~
## $ VA.EST                                                 <dbl> NaN, -1.9~
## $ VC.IHR.PSRC.P5                                         <dbl> NaN, NaN,~
```

```
head(wb_undp_df_r)
```

```
##   ISO3 Country Name year hdi - Human Development Index (value)
## 1  AFG  Afghanistan 1995                                 0.329
## 2  AFG  Afghanistan 1996                                 0.334
## 3  AFG  Afghanistan 1997                                 0.338
## 4  AFG  Afghanistan 1998                                 0.338
## 5  AFG  Afghanistan 1999                                 0.347
## 6  AFG  Afghanistan 2000                                 0.351
```

```
##   gii_rank - GII Rank rankdiff_hdi_phdi - Difference from HDI rank
## 1                NaN                                            NaN
## 2                NaN                                            NaN
## 3                NaN                                            NaN
## 4                NaN                                            NaN
## 5                NaN                                            NaN
## 6                NaN                                            NaN
##   ihdi - Inequality-adjusted Human Development Index (value)
## 1                                                       NaN
## 2                                                       NaN
## 3                                                       NaN
## 4                                                       NaN
## 5                                                       NaN
## 6                                                       NaN
##   coef_ineq - Coefficient of human inequality
## 1                                        NaN
## 2                                        NaN
## 3                                        NaN
## 4                                        NaN
## 5                                        NaN
## 6                                        NaN
##   coef_ineq - Coefficient of human inequality
## 1                                        NaN
## 2                                        NaN
## 3                                        NaN
## 4                                        NaN
## 5                                        NaN
## 6                                        NaN
##   ineq_le - Inequality in life expectancy le - Life Expectancy at Birth (years)
## 1                                    NaN                                 52.103
## 2                                    NaN                                 52.830
## 3                                    NaN                                 53.212
## 4                                    NaN                                 52.487
## 5                                    NaN                                 54.532
## 6                                    NaN                                 55.005
##   ineq_edu - Inequality in eduation ineq_inc - Inequality in income
## 1                              NaN                              NaN
## 2                              NaN                              NaN
## 3                              NaN                              NaN
## 4                              NaN                              NaN
## 5                              NaN                              NaN
## 6                              NaN                              NaN
##   coef_ineq - Coefficient of human inequality
## 1                                        NaN
## 2                                        NaN
## 3                                        NaN
## 4                                        NaN
## 5                                        NaN
## 6                                        NaN
##   coef_ineq - Coefficient of human inequality AG.PRD.FOOD.XD    CC.EST
## 1                                        NaN          67.87       NaN
## 2                                        NaN          71.42 -1.291705
## 3                                        NaN          76.45       NaN
## 4                                        NaN          80.61 -1.176012
```

```
## 5                                         NaN          79.69       NaN
## 6                                         NaN          67.78 -1.271724
##   EG.ELC.ACCS.ZS EN.POP.DNST EN.POP.SLUM.UR.ZS ER.H2O.FWST.ZS FX.OWN.TOTL.ZS
## 1            NaN    26.16536               NaN       59.04386            NaN
## 2            NaN    27.23467               NaN       57.61255            NaN
## 3            NaN    28.29077               NaN       56.18125            NaN
## 4            NaN    29.37613               NaN       54.75702            NaN
## 5            NaN    30.49198               NaN       54.75702            NaN
## 6            4.4    30.86385               NaN       54.75702            NaN
##   GB.XPD.RSDV.GD.ZS    GE.EST IS.SHP.GCNW.XQ IT.NET.BBND.P2 IT.NET.USER.ZS
## 1               NaN       NaN            NaN            NaN            NaN
## 2               NaN -2.175167            NaN            NaN            NaN
## 3               NaN       NaN            NaN            NaN            NaN
## 4               NaN -2.102292            NaN            NaN            NaN
## 5               NaN       NaN            NaN            NaN            NaN
## 6               NaN -2.173946            NaN            NaN            NaN
##   NY.GDP.MKTP.KD NY.GDP.MKTP.KD.ZG NY.GDP.PCAP.KD.ZG NY.GNP.PCAP.KD.ZG
## 1            NaN               NaN               NaN               NaN
## 2            NaN               NaN               NaN               NaN
## 3            NaN               NaN               NaN               NaN
## 4            NaN               NaN               NaN               NaN
## 5            NaN               NaN               NaN               NaN
## 6     6206547590               NaN               NaN               NaN
##      PV.EST    RL.EST SE.ADT.LITR.ZS SE.COM.DURS SE.LPV.PRIM SE.LPV.PRIM.SD
## 1       NaN       NaN            NaN         NaN         NaN            NaN
## 2 -2.417310 -1.788075            NaN         NaN         NaN            NaN
## 3       NaN       NaN            NaN         NaN         NaN            NaN
## 4 -2.427355 -1.734887            NaN           6         NaN            NaN
## 5       NaN       NaN            NaN           6         NaN            NaN
## 6 -2.438969 -1.780661            NaN           6         NaN            NaN
##   SE.SEC.CUAT.LO.ZS SE.SEC.CUAT.PO.ZS SE.SEC.CUAT.UP.ZS SE.TER.CUAT.BA.ZS
## 1               NaN               NaN               NaN               NaN
## 2               NaN               NaN               NaN               NaN
## 3               NaN               NaN               NaN               NaN
## 4               NaN               NaN               NaN               NaN
## 5               NaN               NaN               NaN               NaN
## 6               NaN               NaN               NaN               NaN
##   SE.TER.CUAT.DO.ZS SE.TER.CUAT.MS.ZS SE.TER.CUAT.ST.ZS SE.XPD.TOTL.GD.ZS
## 1               NaN               NaN               NaN               NaN
## 2               NaN               NaN               NaN               NaN
## 3               NaN               NaN               NaN               NaN
## 4               NaN               NaN               NaN               NaN
## 5               NaN               NaN               NaN               NaN
## 6               NaN               NaN               NaN               NaN
##   SH.STA.WASH.P5 SH.UHC.SRVS.CV.XD SH.XPD.CHEX.PC.CD SH.XPD.GHED.PC.CD
## 1            NaN               NaN               NaN               NaN
## 2            NaN               NaN               NaN               NaN
## 3            NaN               NaN               NaN               NaN
## 4            NaN               NaN               NaN               NaN
## 5            NaN               NaN               NaN               NaN
## 6            NaN                23               NaN               NaN
##   SI.POV.GINI SM.POP.TOTL.ZS SN.ITK.MSFI.ZS SP.DYN.LE00.IN SP.RUR.TOTL.ZG
## 1         NaN      0.4264236            NaN         52.103       4.778918
## 2         NaN            NaN            NaN         52.830       3.890502
```

```
## 3         NaN            NaN            NaN         53.212       3.688206
## 4         NaN            NaN            NaN         52.487       3.649521
## 5         NaN            NaN            NaN         54.532       3.611540
## 6         NaN      0.3853275            NaN         55.005       1.094174
##      VA.EST VC.IHR.PSRC.P5
## 1       NaN            NaN
## 2 -1.908540            NaN
## 3       NaN            NaN
## 4 -2.039301            NaN
## 5       NaN            NaN
## 6 -2.031417            NaN
```

```
saveRDS(wb_undp_df_r, file = here::here("Data","PreEDA_DataFrame.rds"))
```